PG No. 14 D_{2d} $\bar{4}2m$ (-42m setting) [tetragonal]

* generator : $2_{001},\ -4^{+}_{001},\ 2_{010}$
* conjugacy class

$[1]$: 1
$[2_{001}]$: 2_{001}
$[2_{100}]$: $2_{100},\ 2_{010}$
$[\mathrm{m}_{110}]$: $\mathrm{m}_{110},\ \mathrm{m}_{1-10}$
$[-4^{+}_{001}]$: $-4^{+}_{001},\ -4^{-}_{001}$

* symmetry operation

Table 1: Symmetry operations for 3d polar vector.

No.	tag	matrix (polar)	det
1	1	$\begin{bmatrix}1&0&0\\0&1&0\\0&0&1\end{bmatrix}$	1
2	2_{001}	$\begin{bmatrix}-1&0&0\\0&-1&0\\0&0&1\end{bmatrix}$	1
3	-4^{+}_{001}	$\begin{bmatrix}0&1&0\\-1&0&0\\0&0&-1\end{bmatrix}$	-1
4	-4^{-}_{001}	$\begin{bmatrix}0&-1&0\\1&0&0\\0&0&-1\end{bmatrix}$	-1
5	2_{010}	$\begin{bmatrix}-1&0&0\\0&1&0\\0&0&-1\end{bmatrix}$	1
6	2_{100}	$\begin{bmatrix}1&0&0\\0&-1&0\\0&0&-1\end{bmatrix}$	1
7	m_{110}	$\begin{bmatrix}0&-1&0\\-1&0&0\\0&0&1\end{bmatrix}$	-1
8	m_{1-10}	$\begin{bmatrix}0&1&0\\1&0&0\\0&0&1\end{bmatrix}$	-1